

IBM Data Science Capstone Project

SpaceX Launch Records Dashboard – Final Report

Objective

Analyze SpaceX Falcon 9 launch data using Dash and Plotly to study launch success by site, payload, and booster version.

Key Findings

KSC LC-39A recorded the largest number of successful launches. KSC LC-39A showed the highest launch success rate. Payloads around 5,000–9,500 kg achieved the highest success rate. Payloads below about 2,000 kg showed the lowest success rate. Booster Version B5 demonstrated the highest launch success rate, followed by FT.

Conclusion

The dashboard demonstrates that newer Falcon 9 booster versions and medium-to-heavy payloads are associated with higher launch success. Interactive filtering enables efficient exploration of launch site performance, payload effects, and booster reliability.